

COL362/7362/632 Introduction to Data Management

Relational Algebra

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Why Relational Algebra?

- ▶ Mathematical core of the relational model
- ▶ Relational Algebra allows
 - Manipulating data declaratively
 - Composing complex queries from simple building blocks
 - Systems to reason about queries without executing them
- ▶ RA has survived for over 50 years, the hidden power behind every modern data system

Why Abstraction Matters?

Abstraction lets us **separate what we want from how it is done**

- ▶ You describe what data you want
- ▶ The system decides how to get it efficiently

Relational Algebra is a **clean, minimal abstraction**:

- ▶ No indexes
- ▶ No memory layout
- ▶ No parallelism
- ▶ No machines

Just **sets (or multisets) of tuples and operators over them**. This separation is what allows:

- ▶ Query optimization
- ▶ Portability across systems
- ▶ Execution on laptops, clusters, or the cloud
- ▶ Evolution from single-machine Data Systems to Big Data systems like Spark, Flink, and to ML systems like Tensorflow, etc.

Relational Algebra

- ▶ If R is a relation, then

$$R' := \langle op \rangle_{\langle param \rangle}(R) \text{ is also a relation}$$

- ▶ If R_1 and R_2 are relations, then

$$R' := R_1 \langle op \rangle_{\langle param \rangle} R_2 \text{ is also a relation}$$

- ▶ Relational algebra is a procedural query language comprising relational algebra operators
- ▶ Operates on relations and produces relations
- ▶ Basis of many data processing systems — provides a simple and effective abstraction for querying and data manipulation

Selection σ

Filter tuples based on a condition

$$R' := \sigma_{\theta}(R) \subseteq R$$

$$\triangleright \theta : [R] \mapsto \{ \text{true}, \text{false} \}$$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

$\sigma_{\text{position} > 5}(\text{Player})$		
player_id	name	position
123	MS Dhoni	7
142	Ashok Singh	8
163	Ashok Singh	6

$\sigma_{\text{name} = \text{'Ashok Singh'}}(\text{Player})$		
player_id	name	position
142	Ashok Singh	8
163	Ashok Singh	6

$\sigma_{\text{position} \leq 7 \wedge \text{player_id} < 150}(\text{Player})$		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1

Projection π

Retrieves specific attributes

$$R' := \pi_{[R']}(R)$$

$$\triangleright [R'] \subseteq [R]$$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

$\pi_{id}(\text{Player})$
player_id
123
134
131
142
154
163
250

$\pi_{name}(\text{Player})$
name
MS Dhoni
SK Yadav
Rohit Sharma
Ashok Singh
David Warner
Chris Gayle

$\pi_{name, position}(\text{Player})$	
name	position
MS Dhoni	7
SK Yadav	5
Rohit Sharma	1
Ashok Singh	8
David Warner	3
Ashok Singh	6
Chris Gayle	1

Union $\cup$

Combines tuples from two relations

$$R' := R_1 \cup R_2$$

► $[R'] := [R_1] = [R_2]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

MIPlayer		
player_id	name	position
134	SK Yadav	5
131	Rohit Sharma	1
176	Hardik Pandaya	5

Player $\cup$ MIPlayer		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1
176	Hardik Pandaya	5

Difference —

Finds tuples in one relation but not the other

$$R' := R_1 - R_2$$

► $[R'] := [R_1] - [R_2]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

MIPlayer		
player_id	name	position
134	SK Yadav	5
131	Rohit Sharma	1
176	Hardik Pandaya	5

Player - MIPlayer		
player_id	name	position
123	MS Dhoni	7
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Crossproduct $\times$

Combines tuples from two relations

$$R' := R_1 \times R_2$$

- ▶ $[R'] := [R_1] \cup [R_2]$
- ▶ $|R'| = |R_1| \cdot |R_2|$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

League	
id	league_name
11	IPL
21	CPL

Player $\times$ League				
player_id	name	position	id	league_name
123	MS Dhoni	7	11	IPL
123	MS Dhoni	7	12	CPL
134	SK Yadav	5	11	IPL
134	SK Yadav	5	12	CPL
131	Rohit Sharma	1	11	IPL
131	Rohit Sharma	1	12	CPL
142	Ashok Singh	8	11	IPL
142	Ashok Singh	8	12	CPL
154	David Warner	3	11	IPL
154	David Warner	3	12	CPL
163	Ashok Singh	6	11	IPL
163	Ashok Singh	6	12	CPL
250	Chris Gayle	1	11	IPL
250	Chris Gayle	1	12	CPL

Rename ρ

Allows renaming of a relation or its attributes

$$R' := \rho'_R(R)$$

► $[R'] := [R]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

$\rho_{\text{cricketers}}(\text{Player})$		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

$$R' := \rho_{A' \leftarrow A}(R)$$

► $A \in R$

$\rho_{\text{position} \leftarrow \text{b_order}}(\text{Player})$		
player_id	name	b_order
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Intersection $\cap$

$$R' := R_1 \cap R_2$$

$$\triangleright = R_1 - (R_1 - R_2)$$

$$\triangleright [R'] := [R_1] = [R_2]$$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Player $\cap$ MIPlayer		
player_id	name	position
134	SK Yadav	5
131	Rohit Sharma	1

MIPlayer		
player_id	name	position
134	SK Yadav	5
131	Rohit Sharma	1
176	Hardik Pandaya	5

Theta join $\bowtie_{\theta}$

$$R' := R_1 \bowtie_{\theta} R_2$$

- ▶ $= \sigma_{\theta}(R_1 \times R_2) \subseteq R_1 \times R_2$
- ▶ $[R'] := [R_1] \cup [R_2]$
- ▶ $\theta : [R_1] \cup [R_2] \mapsto \{\text{true}, \text{false}\}$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

League	
id	league_name
150	IPL
210	CPL

Player $\bowtie_{\theta:=\text{true}}$ League				
player_id	name	position	id	league_name
123	MS Dhoni	7	150	IPL
123	MS Dhoni	7	210	CPL
134	SK Yadav	5	150	IPL
134	SK Yadav	5	210	CPL
131	Rohit Sharma	1	150	IPL
131	Rohit Sharma	1	210	CPL
142	Ashok Singh	8	150	IPL
142	Ashok Singh	8	210	CPL
154	David Warner	3	150	IPL
154	David Warner	3	210	CPL
163	Ashok Singh	6	150	IPL
163	Ashok Singh	6	210	CPL
250	Chris Gayle	1	150	IPL
250	Chris Gayle	1	210	CPL

Theta join $\bowtie_{\theta}$ with selective predicate

$$R' := R_1 \bowtie_{\theta} R_2$$

- ▶ $= \sigma_{\theta}(R_1 \times R_2) \subseteq R_1 \times R_2$
- ▶ $[R'] := [R_1] \cup [R_2]$
- ▶ $\theta : [R_1] \cup [R_2] \mapsto \{ \text{true}, \text{false} \}$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

League	
id	league_name
150	IPL
210	CPL

Player $\bowtie_{\theta=\text{player.id}>\text{id}}$ League				
player_id	name	position	id	league_name
154	David Warner	3	150	IPL
163	Ashok Singh	6	150	IPL
250	Chris Gayle	1	150	IPL
250	Chris Gayle	1	210	CPL

Equi join $\bowtie$

$$R' := R_1 \bowtie_{[L][R]} R_2$$

$$:= R_1 \bowtie_{\theta} R_2$$

$$\triangleright \theta := R_1.[L] == R_2.[R] \text{ and } [L] \subseteq [R_1], [R] \subseteq [R_2]$$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Batter	
id	bt_type
123	finisher
131	opener
250	opener
272	lower order

Player $\bowtie_{\text{player.id, id}}$ Batter				
player_id	name	position	id	bt_type
123	MS Dhoni	7	123	finisher
131	Rohit Sharma	1	131	opener
250	Chris Gayle	1	250	opener

Natural join $\bowtie$

$$R' := R_1 \bowtie_{[J][J]} R_2$$

$$:= \pi_{[R']} (R_1 \bowtie_{[J][J]} R_2)$$

► $[J] := [R_1] \cap [R_2]$ and $[R'] := [R_1] \cup ([R_2] - [J])$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Player $\bowtie$ Batter			
player_id	name	position	bt_type
123	MS Dhoni	7	finisher
131	Rohit Sharma	1	opener
250	Chris Gayle	1	opener

Batter	
player_id	bt_type
123	finisher
131	opener
250	opener

Left semi-join $\bowtie$

$$R' := R_1 \bowtie R_2$$

$$:= \pi_{[R_1]}(R_1 \bowtie R_2) \subseteq R_1$$

► $[R'] := [R_1]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Batter	
player_id	bt_type
123	finisher
131	opener
250	opener

Player $\bowtie$ Batter		
player_id	name	position
123	MS Dhoni	7
131	Rohit Sharma	1
250	Chris Gayle	1

Right semi-join $\bowtie$

$$R' := R_1 \bowtie R_2$$

$$:= \pi_{[R_2]}(R_1 \bowtie R_2) \subseteq R_2$$

► $[R'] := [R_2]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Batter	
player_id	bt_type
123	finisher
131	opener
267	middle order

Player $\bowtie$ Batter	
player_id	bt_type
123	finisher
131	opener

Left anti semi-join $\triangleright$

$$\begin{aligned} R' &:= R_1 \triangleright R_2 \\ &= R_1 - (R_1 \bowtie R_2) \subseteq R_1 \end{aligned}$$

► $[R'] := [R_1]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Batter	
player_id	bt_type
123	finisher
131	opener
267	middle order

Player $\triangleright$ Batter		
player_id	name	position
134	SK Yadav	5
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Right anti semi-join $\triangleleft$

$$\begin{aligned} R' &:= R_1 \triangleleft R_2 \\ &= R_2 - (R_1 \bowtie R_2) \subseteq R_2 \end{aligned}$$

► $[R'] := [R_2]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Batter	
player_id	bt_type
123	finisher
131	opener
267	middle order

Player $\triangleleft$ Batter	
player_id	bt_type
267	middle order

Left outer join $\bowtie$

$$R' := R_1 \bowtie R_2$$

$$= (R_1 \bowtie R_2) \cup ((R_1 \triangleright R_2) \times R_3)$$

- ▶ where $R_3 = \{(\omega, \dots, \omega)\}$ and $|(\omega, \dots, \omega)| = |[R'] - [R_1]|$
- ▶ $[R'] := [R_1 \bowtie R_2]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Batter	
player_id	bt_type
123	finisher
131	opener
267	middle order

Player $\bowtie$ Batter			
player_id	name	position	bt_type
123	MS Dhoni	7	finisher
134	SK Yadav	5	ω
131	Rohit Sharma	1	opener
142	Ashok Singh	8	ω
154	David Warner	3	ω
163	Ashok Singh	6	ω
250	Chris Gayle	1	ω

Right outer join $\bowtie\lrcorner$

$$R' := R_1 \bowtie\lrcorner R_2$$

$$= (R_1 \bowtie R_2) \cup (R_3 \times (R_1 \triangleleft R_2))$$

- ▶ where $R_3 = \{(\omega, \dots, \omega)\}$ and $|(\omega, \dots, \omega)| = |[R'] - [R_2]|$
- ▶ $[R'] := [R_1 \bowtie R_2]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Batter	
player_id	bt_type
123	finisher
131	opener
267	middle order

Player $\bowtie\lrcorner$ Batter			
player_id	name	position	bt_type
123	MS Dhoni	7	finisher
131	Rohit Sharma	1	opener
267	ω	ω	middle order

Outer join $\bowtie$

$$\begin{aligned} R' &:= R_1 \bowtie R_2 \\ &= R_1 \bowtie R_2 \cup R_1 \ltimes R_2 \end{aligned}$$

► $[R'] := [R_1 \ltimes R_2]$

Player		
player_id	name	position
123	MS Dhoni	7
134	SK Yadav	5
131	Rohit Sharma	1
142	Ashok Singh	8
154	David Warner	3
163	Ashok Singh	6
250	Chris Gayle	1

Batter	
player_id	bt_type
123	finisher
131	opener
267	middle order

Player $\bowtie$ Batter			
player_id	name	position	bt_type
123	MS Dhoni	7	finisher
134	SK Yadav	5	ω
131	Rohit Sharma	1	opener
142	Ashok Singh	8	ω
154	David Warner	3	ω
163	Ashok Singh	6	ω
250	Chris Gayle	1	ω
267	ω	ω	middle order

Grouping and aggregation Γ

$$R' := \Gamma_{[G], f_1(G_1), \dots, f_n(G_n)}(R)$$

- ▶ $[G] \subseteq [R]$ and $[G_i] \subset [R]$
- ▶ $f_i : [G_i] \mapsto D_i$

Common aggregate functions

- ▶ sum
- ▶ min
- ▶ max
- ▶ avg
- ▶ count
- ▶ count-distinct*

Player			
player_id	name	team	runs
123	MS Dhoni	CSK	600
134	SK Yadav	MI	400
131	Rohit Sharma	MI	500
142	Ashok Singh	CSK	100
154	David Warner	DC	150
163	Ashok Singh	CSK	250
250	Chris Gayle	RCB	300

$\Gamma_{team, \text{sum}(\text{runs})}(\text{Player})$	
team	sum(runs)
CSK	950
MI	900
DC	150
RCB	300

Generalized projections

$$R' := \pi_{E_1, \dots, E_n}(R)$$

- ▶ E_i is arithmetic expression involving constants and attributes of R
- ▶ Same as projection, but some operation can be “directly” expressed

Player			
player_id	name	bat_first	bat_second
123	MS Dhoni	1000	600
134	SK Yadav	900	400
131	Rohit Sharma	800	500
142	Ashok Singh	300	100
154	David Warner	800	150
163	Ashok Singh	100	250
250	Chris Gayle	1200	300

$\pi_{\text{name, bat_first} + \text{bat_second}}(\text{Player})$	
name	bat_first + bat_second
MS Dhoni	1600
SK Yadav	1300
Rohit Sharma	1300
Ashok Singh	400
David Warner	950
Ashok Singh	350
Chris Gayle	1500

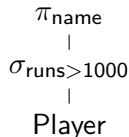
More operators

- ▶ duplicate elimination ($\delta(R)$)
- ▶ Sorting τ
- ▶ More on this later

Composing Operators

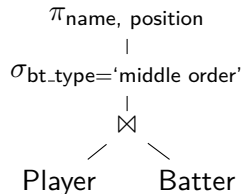
- ▶ Names of players with more than 1000 runs

$\pi_{\text{name}}(\sigma_{\text{runs} > 1000}(\text{Player}))$



- ▶ Batters and their position who play in the middle order

$\pi_{\text{name}, \text{position}}(\sigma_{\text{bt_type} = \text{'middle order'}}(\text{Player} \bowtie \text{Batter}))$



Examples

Students		
StudentID	Name	Major
1	Akash	CS
2	Binoy	Math
3	Charu	CS

Courses		
CourseID	CourseName	InstructorID
101	Databases	201
102	Calculus	202
103	Algorithms	201

Instructors		
InstructorID	Name	Department
201	Prof. Neetu	CS
202	Prof. Ram	Math

- Retrieve names of all students majoring in “CS”

$$\pi_{name}(\sigma_{major='CS'}(Students))$$

- Find instructors who teach courses in both CS and Math departments

$$R = \pi_{InstructorID}(\sigma_{Department='CS'}(Instructors))$$

$$S = \pi_{InstructorID}(\sigma_{Department='Math'}(Instructors))$$

$$R \cap S$$

Examples

Students		
StudentID	Name	Major
1	Akash	CS
2	Binoy	Math
3	Charu	CS

Courses		
CourseID	CourseName	InstructorID
101	Databases	201
102	Calculus	202
103	Algorithms	201

Instructors		
InstructorID	Name	Department
201	Prof. Neetu	CS
202	Prof. Ram	Math

Enrollments		
EnrollmentID	StudentID	CourseID
1	1	101
2	2	102
3	3	101
4	1	103
5	3	103

- Find all students enrolled in courses taught by "Prof. Neetu."

1. Find the 'InstructorID' of "Prof. Neetu":

$$R_1 = \sigma_{\text{Name}='Prof. Neetu'}(\text{Instructors})$$

2. Join this with the 'Courses' table to find the courses taught by "Prof. Neetu":

$$R_2 = R_1 \bowtie_{\text{Instructors.InstructorID}=\text{Courses.InstructorID}} \text{Courses}$$

3. Join the result with the 'Enrollments' table:

$$R_3 = R_2 \bowtie_{\text{Courses.CourseID}=\text{Enrollments.CourseID}} \text{Enrollments}$$

4. Retrieve the names of the students:

$$\pi_{\text{Name}}(R_3 \bowtie_{\text{Students.StudentID}=\text{Enrollments.StudentID}} \text{Students})$$

Examples

Students		
StudentID	Name	Major
1	Akash	CS
2	Binoy	Math
3	Charu	CS

Courses		
CourseID	CourseName	InstructorID
101	Databases	201
102	Calculus	202
103	Algorithms	201

Instructors		
InstructorID	Name	Department
201	Prof. Neetu	CS
202	Prof. Ram	Math

Enrollments		
EnrollmentID	StudentID	CourseID
1	1	101
2	2	102
3	3	101
4	1	103
5	3	103

- List students who are not taking any courses.

1. Find all 'StudentID' who are enrolled in courses:

$$R_1 = \pi_{\text{StudentID}}(\text{Enrollments})$$

2. Find all 'StudentID' in the 'Students' table:

$$R_2 = \pi_{\text{StudentID}}(\text{Students})$$

3. Use set difference to find students not enrolled in any course:

$$R_2 - R_1$$

Homework

Students		
StudentID	Name	Major
1	Akash	CS
2	Binoy	Math
3	Charu	CS

Courses		
CourseID	CourseName	InstructorID
101	Databases	201
102	Calculus	202
103	Algorithms	201

Instructors		
InstructorID	Name	Department
201	Prof. Neetu	CS
202	Prof. Ram	Math

Enrollments		
EnrollmentID	StudentID	CourseID
1	1	101
2	2	102
3	3	101
4	1	103
5	3	103

- ▶ Find all courses taken by every CS major.
- ▶ Find instructors who teach at least two different courses.
- ▶ Find pairs of students who are enrolled in the same course.

Summary

- ▶ Relational algebra is a foundational tool for understanding and building queries
 - Basis of SQL
 - Basis of modern data processing systems
- ▶ Key operations
 - Selection, projection, union, set difference, cross product, joins
- ▶ Relational Algebra isn't just theory. **Its everywhere:**
 - Relational DBs (PostgreSQL, Oracle, MySQL, ...)
 - Big Data Systems and (Flink, Presto, Spark, ...)
 - Graph Systems
 - Streaming Systems
 - Machine Learning feature pipelines and systems

Next Lecture

- ▶ SQL Hands-on sessions
- ▶ Before coming to class
 - Install PostgreSQL
 - Make sure you can run Jupyter notebooks